

AI Usage Patterns & Prompt Quality Analysis

An exploratory analysis of user behaviour, prompt efficiency, AI dependency, and student AI usage
Then a short description:

This report analyses AI interaction patterns across multiple datasets to understand how users engage with AI systems. The analysis explores prompt quality, user behaviour, confidence, AI dependency, and student usage patterns to identify meaningful insights and support more effective and responsible AI use.

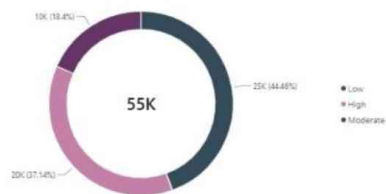
Data sources:

- LMSYS
- Open Assistant (OASST)
- Student Survey Data
- Chat GPT

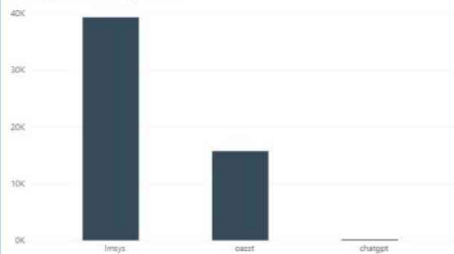


Prompt Analysis Dashboard

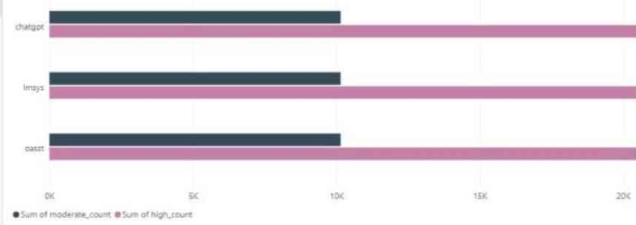
Prompt Label Distribution



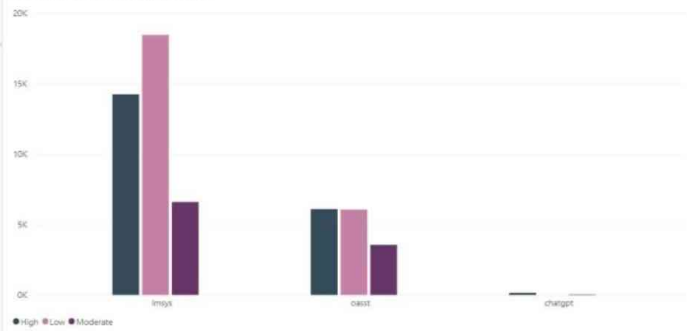
Prompt Volume By Source



Prompt Efficiency Breakdown



Prompt Label Mix By Source



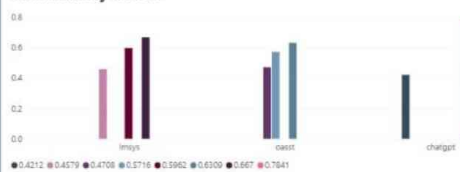
Total Prompts	High Efficiency Prompts	Total Users
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Sum of value
55.18K

Sum of value
20.50K

Total Users
21.72K

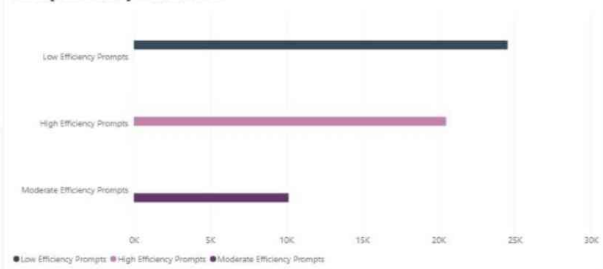
Confidence By Source



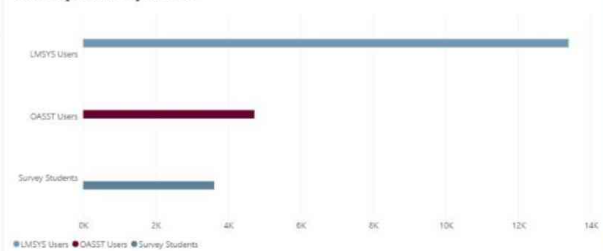
Executive Metrics Summary

Category	Metric	Sum of Value
Model Performance	Average Confidence	0.59
Prompt Efficiency	High Efficiency Prompts	20,499.00
Prompt Efficiency	Low Efficiency Prompts	24,531.00
Prompt Efficiency	Moderate Efficiency Prompts	10,151.00
Prompt Volume	Total Prompts	\$5,177.00
User Population	LMSYS Users	13,363.00
User Population	OASST Users	4,721.00
User Population	Survey Students	3,614.00
Total		1,32,072.59

Prompt Efficiency Distribution

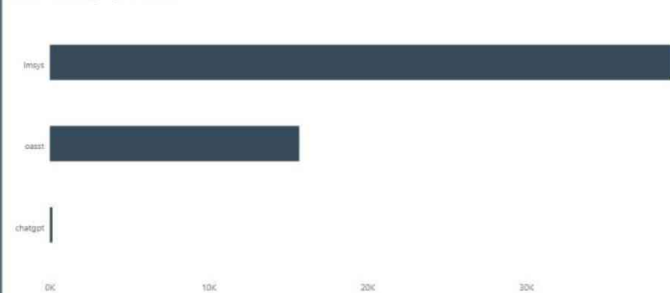


User Population By Source

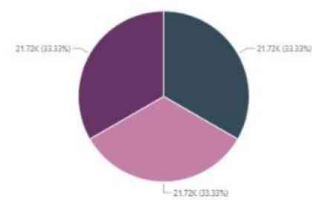


User Behavior Analysis

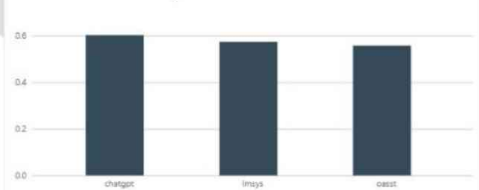
User Activity by source



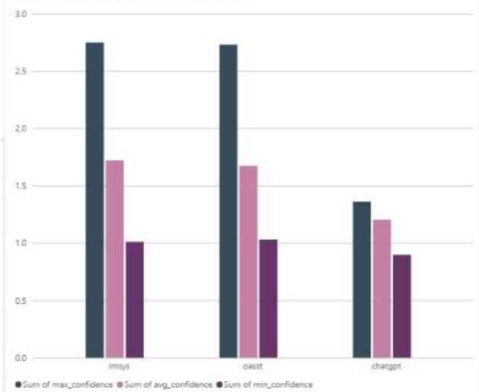
User Distribution By Source



Confidence Variation By Source



Confidence Range Across Sources



AI Dependency Analysis

Dependency Tier Distribution



LMSYS User Count

Sum of user_count
13K

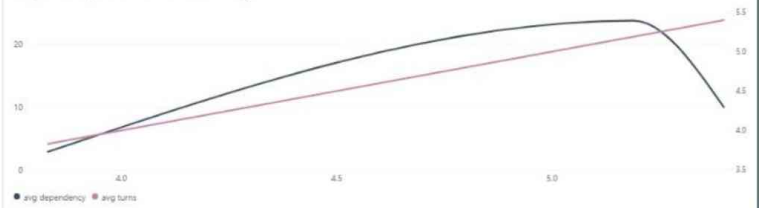
OASST User Count

Sum of user_count
5K

Avg Prompts Per User (Overall)

Average Prompts per User
2.54

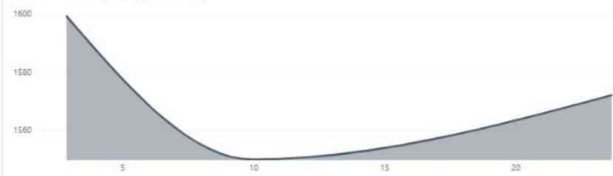
Dependency vs Conversation Length



LMSYS Average Dependency



OASST Average Dependency



Student AI Usage Analysis

Total Students

Sum of student_count
4K

Academic AI Users

Sum of academic_users
1K

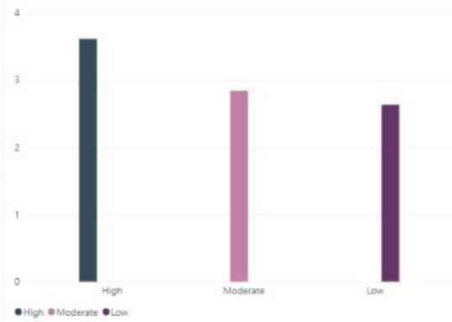
Avg Daily AI Usage

Average of avg_daily_usage_hours
2.56

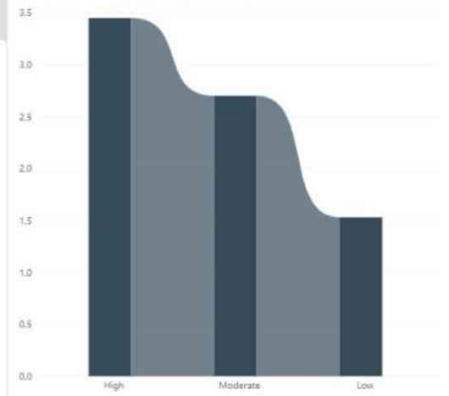
Avg AI Tools Used

Sum of avg_ai_tools_used
4.44

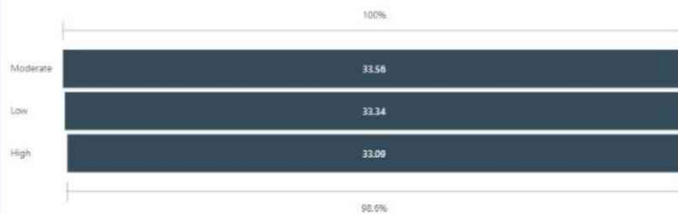
AI Trust By Dependency Tier



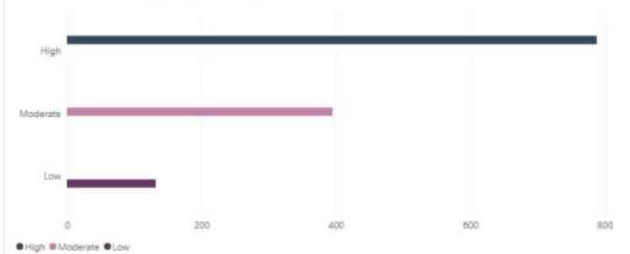
Daily AI Usage By Dependency Tier



Student Dependency Distribution



Academic Users By Dependency Tier

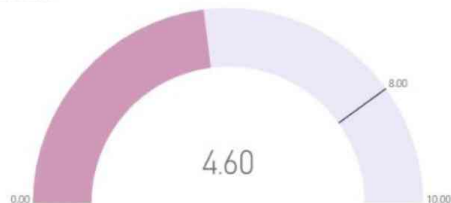


Recommendations And Insights

High Efficiency Prompts	Avg LMSYS Dependency	Avg OASST Dependency	Avg Daily Student AI Usage
High Efficiency Prompts 20.50K	Average LMSYS Dependency 18.10	Average OASST Dependency 12.16	Average Daily Student AI Usage (in hrs) 2.56

The analysis suggests that the next stage of AI adoption should focus on better usage rather than simply greater usage. Users and students would benefit from improved AI literacy, including training on effective prompting, evaluating AI-generated information, recognizing limitations, and maintaining independent decision making.

Avg Confidence



Recommendation 1

Improve Prompt Literacy

A large share of interactions fall outside the high-efficiency category. Teaching users how to provide clear context, specific instructions, and structured prompts could significantly improve AI output quality.

Recommendation 2

Encourage Critical AI Use

Users should treat AI-generated responses as assistance rather than unquestionable truth. Verification, fact-checking, and independent reasoning remain essential, particularly in academic and professional contexts.

Recommendation 3

Monitor Excessive Dependency

Higher usage and trust may be associated with stronger reliance on AI. Educational institutions should encourage balanced AI use that supports learning without replacing core problem-solving and critical-thinking skills.

Recommendation 4

Promote Responsible Academic AI Adoption

Students should be encouraged to use AI for brainstorming, explanation, research assistance, and productivity while maintaining transparency and academic integrity.